

WILLIAM FORD

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ABOUT ME

Originally from England, I am a second-year mathematics PhD candidate jointly at École Polytechnique and Université Paris-Saclay. My research focuses on optimal transport, with applications in machine learning, computer vision, statistics and numerical analysis. I wrote two sole-author preprints in my first PhD year, ranked **1st of 25** in my MSc, achieved an M2 average of **18.1/20**, and ranked **7th of 129** in my BSc.

EDUCATION

CMAP, École Polytechnique / LMO, Université Paris-Saclay – PhD Mathematics 2025–2028

Research on quantitative stability of optimal transport. Supervisors: *Dr Michael Goldman and Dr Cyril Letrouit*.

Institut Polytechnique de Paris PhD Track / Paris-Saclay M2 Optimisation 2024–2025

Très Bien, 18.1/20. Stochastic optimisation, nonsmooth analysis, optimal control, optimal transport, game theory.

Durham University – MSc Mathematics 2023–2024

First Class, 88/100; ranked 1st of 25. Analysis, partial differential equations, stochastic analysis, ergodic theory.

Durham University – BSc Mathematics 2020–2023

First Class, 82/100; ranked 7th of 129. Analysis, probability, statistics, data science, abstract algebra, number theory.

RESEARCH PROJECTS

My current research is on the stability of the optimal transport problem. Before my PhD, this was very poorly understood, despite being crucial to applications in machine learning and statistics, where it is important to quantify errors introduced by various approximations and estimations.

Quantitative Stability, Coercivity and Uniqueness of Optimal Transport Plans – preprint Sept. 2026

In my most recent paper, I prove that when *one* of the marginal probabilities is suitably regular, the optimal transport problem is locally quantitatively stable with respect to perturbation of *both* marginals. My results also have direct implications for the statistical estimation of transport maps.

Quantitative Uniqueness of Kantorovich Potentials – submitted Apr. 2026

In my first paper, I gave the first quantitative bounds for almost-uniqueness of the dual optimal transport problem, as well as vastly generalising the sharp hypotheses for uniqueness to allow for lower-dimensional marginal measures. This was a result of a novel geometric insight regarding the structure of the set of optimal potentials.

Quantitative Stability in Discrete Optimal Transport – M2 research thesis Oct. 2025

During my M2 research internship, I studied the stability properties of the fully discrete optimal transport problem. This has intimate connections with linear programming, and taking this novel approach to stability led to the uniqueness results in my first paper.

QUANTITATIVE AND TECHNICAL SKILLS

Mathematics	Probability, stochastic analysis, optimisation, statistics, optimal control, partial differential equations, ergodic theory, mathematical finance, linear algebra, calculus, numerical analysis.
Programming	Python (NumPy, pandas, PyTorch, Matplotlib), AI-assisted development, agentic workflows, LaTeX.
Spoken languages	English (native), French (C1), German (A2), Italian (A1, <i>ma sto imparando</i>).
Interests	Road and mountain biking, bikepacking, swimming, classical & jazz trumpet, close-up magic.

AWARDS AND SCHOLARSHIPS

2024	Sophie Germain M2 Excellence Scholarship (€11,000), Fondation Mathématique Jacques Hadamard.
2024	Full M2 and PhD Track scholarship, Institut Polytechnique de Paris.
2024	Winton MSc Memorial Prize (£500), highest-scoring Mathematics MSc student, Durham University.
2021	Norton Prize, performance in science, St Cuthbert's Society, Durham University.

WORK EXPERIENCE

Teaching Assistant – École Polytechnique	2025–2026
Tutorial Leader and Teaching Assistant – Durham University	2023–2024
COVID-19 Support – Durham University	2021–2022
Online Tutor – MyTutor	2021